

Assessment 2 Specification

Read this assessment brief carefully, it tells you how you are going to be assessed, how to submit your assessment on-time and how (and when) you'll receive your marks and feedback.

Module Code	CSI_5_ISM
Module Title	Information Systems Management
Lecturer	Francis Babayemi - Module Leader
% of Module Mark	Marked 100%
Distributed	Monday 17 November 2025
Submission Method	• Submit online via this Module's Moodle site • Submit a single 2,500 word PDF document (±10%) • Marked 100%
Submission Deadline	Friday 19 December 2025 - 5pm
Release of Feedback & Marks	Feedback and provisional marks will be available in the Gradebook on Moodle from 09 January 2026

Assessment Aim:

The aim of this individual assignment is to assess students' ability to review, reflect, and propose an IS solution based on a business case. Students will need to demonstrate that they understand the role of computer-based ISs and how they support the business processes within an organisation. Also, the assessment tests students ability to analyse organisational problems and engage in critical evaluation of ISs effectiveness and solution elicitation.

Coursework Details:

Type:	Individual case-based IS evaluation assessment
Assessment requirements	This individual assignment requires you to engage deeply with the case study “Nexora Group’s Digital Rebirth – A Lifecycle-Led Transformation”. You must <u>select one (1) essay question</u> from the nine listed below and produce a 2,500-word essay (±10%) . Your essay must demonstrate a critical and applied understanding of Information Systems Management (ISM), with specific reference to the Information Systems Life Cycle (ISLC) framework. Essay structure/assessment criteria. You are expected to: 1. Contextualise the challenge – Analyse the Nexora case study to identify the relevant IS problem and explain why it is significant. 2. Identify and frame the challenge – Define the technological or organisational challenge in clear ISM terms. 3. Conduct evidence-based research – Draw upon recent academic literature, industry reports, and case studies to evaluate how organisations are addressing similar challenges. 4. Apply IS frameworks – Use the ISLC (and other relevant IS theories) to structure your analysis of how systems should be designed, developed, implemented, and evaluated. 5. Critically assess solutions – Propose realistic solutions to Nexora’s challenge, supported by research. Assess the opportunities and risks of adopting the technology, including governance, infrastructure, and cultural readiness. You must include at least one meaningful system architecture diagram or illustration to demonstrate your understanding of IS design. 6. Acknowledge limitations – Discuss the risks, constraints, or trade-offs in your proposed solution. 7. Demonstrate depth and professionalism – Your essay must demonstrate depth of analysis, integration of theory and practice, and professional presentation. You are expected to uphold academic integrity with correct Harvard referencing, error-free writing, and a coherent, well-structured narrative. NOTE: Your essay must have a Cover Page and it must include your student's credentials, module code, and submission date. Assessment focus • Word Count: 2,500 words (±10%, excluding references, appendices, and figures). • Frameworks: ISLC must be applied throughout. Other IS theories may be used where relevant. • Illustrations: At least one system architecture diagram or illustration must be included to demonstrate system design considerations. • Sources: Minimum of 10–12 academic references, plus industry case studies and reports. • Referencing: Harvard style throughout. • Style: Clear, professional, and analytical, with error-free grammar and logical flow.

Case study	Nexora Group's Digital Rebirth – A Lifecycle-Led Transformation In the autumn of 2022, the boardroom at Nexora Group felt less like the helm of a global logistics leader and more like a high-stakes episode of Survivor: Enterprise Edition. Slide after slide in the quarterly strategy review showed a company losing ground—losing speed—while smaller, more agile competitors were eating its lunch and tweeting about it in real-time. Once an industry titan with operations in 42 countries and a history of innovation in supply chain technology, Nexora had become bogged down by its own weight: ageing infrastructure, fractured systems, compliance headaches, and employee grumbles loud enough to echo across time zones. But the most sobering slide of all wasn't the financial forecast—it was the IT systems architecture map. Half the board didn't understand it; the other half didn't believe it. Hundreds of spaghetti-like connections between 17-year-old servers, outdated mobile apps, siloed databases, and an ERP system that still used Internet Explorer as its default browser. As the CTO muttered, "We're a 21st-century company running on dial-up logic," the CFO, without humour, asked if that was a joke. It wasn't. This, then, was the beginning of Project Helix—a full-spectrum digital rebirth, anchored firmly in the Information Systems Life Cycle (ISLC) and designed not to patch Nexora's problems but to reinvent the organisation from its foundations. The Planning phase was equal parts technical audit and therapy session. Teams across Nexora were interviewed—not just about systems but about frustrations, fears, and even a few confessions ("I've been printing invoices and scanning them back in because the CRM crashes after 4pm," one sales rep admitted). Amidst the chaos, patterns emerged. AI prototypes—some brilliant—were locked away in sandbox environments, untrusted and unused. Customer service was logging complaints on spreadsheets saved to desktops titled "final_version_final_v2.xls". Mobile app reviews read like roast comedy nights. And worst of all, nobody owned the data—everyone just blamed someone else. The ISLC Planning stage uncovered not just gaps but also opportunities: a workforce hungry for change, a leadership team willing to bet on transformation, and an existential threat strong enough to motivate the unmovable. The first order of business? Infrastructure. Nexora's on-premises systems were like that one old relative who refuses to fly but still wants to visit five countries. Frequent outages during peak demand periods had become the punchline to internal jokes ("It's 11:30—time for the ERP to crash again!"). The Design and Development phases initiated a hybrid cloud migration, focusing on modular microservices and Zero Trust Architecture. The VPN was retired ceremoniously (with an actual office party involving a cardboard tombstone labelled "VPN: 2006–2022"). Containerisation through Kubernetes was rolled out; Azure provided regional compliance support. Legacy databases were decoupled from front-end systems like a toxic relationship finally severed. Results came quickly. Downtime dropped by 91%, and customer churn due to reliability issues halved in less than a year. However, technical debt did not go quietly. Hidden hard-coded integrations and one "forgotten" system hosted on a developer's home server (yes, really) reminded the team that ghosts from the past can linger in the most surprising ways. With infrastructure modernised, Nexora turned to its crown jewel: generative AI. In a twist of irony, the company had developed a sophisticated AI model for predictive logistics three years
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prior—RouteGen—but abandoned it due to “lack of readiness.” It had sat idle, like a trained pilot waiting for someone to build the plane.

This time, the team moved deliberately through ISLC stages. Governance came first: an AI Ethics Council was formed, drawing lessons from ISO/IEC 42001. Then came Development, with RouteGen integrated into production via MLOps pipelines. The real triumph, however, was cultural: AI wasn’t implemented to replace planners but to augment them.

RouteGen suggested optimal routes based on weather, customs queues, and traffic. Human planners reviewed, adjusted, and fed outcomes back into the system. The result? A 24% improvement in delivery punctuality, with planners reporting they felt “like chess masters with a very smart assistant—not pieces being replaced.”

Of course, not everything went smoothly. At one point, RouteGen mistakenly interpreted “fastest delivery” to mean “including helicopters,” prompting a surprise internal bill for £1.2 million in misclassified logistics options. Lesson learned: guardrails matter.

AI success demanded reliable data—but Nexora’s data environment resembled the Wild West. Departments hoarded their own spreadsheets, shared only via email, and defined “on-time delivery” at least four different ways. Enter the Data Mesh. By redesigning around domain-owned data products, Nexora decentralised responsibility while standardising governance. Apache Kafka and Flink enabled real-time data streaming. Business users gained access to analytics tools without needing to file an IT ticket and wait three weeks.

The transition wasn’t painless. Some managers resisted, asking, “But who do we blame if it goes wrong now?” But as operations improved—reducing decision latency by 61%—resistance gave way to appreciation. Notably, finance reported forecasting accuracy gains of 36%, and warehouse reordering became automatic and context-aware. This seamless flow of information set the stage for smarter human-machine collaboration across the enterprise.

In Nexora’s Singapore fulfilment centre, workers began interacting with cobots—collaborative robots that didn’t replace them, but worked beside them. Through multimodal interfaces (AR glasses, voice commands), workers guided bots through tasks, while bots offered real-time alerts about misplaced inventory or route congestion.

Implementation of these systems revealed both productivity and empathy gains. Employee surveys found that 82% “preferred working with bots” because it made them feel more supported. One worker even named his assigned cobot “Wall-E” and dressed it with a holiday hat during festive seasons. From an ISLC perspective, the symbiotic design was critical. Unlike the AI implementations of some competitors that led to job cuts, Nexora’s strategy focused on augmentation. Productivity rose, error rates fell, and employee retention in high-turnover roles improved for the first time in four years.

Operating globally, Nexora faced a compliance minefield: GDPR in Europe, India’s Data Protection Bill, and DORA in finance-linked contracts. Initial ISLC planning flagged over 14 data flows that violated regional residency laws. Worse, two data centres were operating under expired security certificates—a fact discovered accidentally by a curious intern. A Governance overhaul followed. Sovereign cloud instances were spun up for sensitive regions. Azure’s confidential computing enabled encrypted processing. DPIAs were rewritten, automated audit trails implemented, and a central compliance command centre launched.

Unexpectedly, this compliance push also unlocked new business. One EU client selected Nexora over competitors due to its transparent DORA-readiness—a reminder that good governance can be a competitive advantage, not just a legal necessity.

Nexora's mobile app journey was a masterclass in humility. Pre-transformation, the app's most-used feature was "close app," and users often preferred calling customer service rather than using the self-service portal. During ISLC Design, UX researchers sat with real users—from truck drivers in rural Brazil to procurement managers in Berlin. They observed everything: slow load times, non-intuitive menus, and a bizarre bug where the app played Christmas music in July.

The redesigned app—built using Flutter for cross-platform consistency—prioritised offline capabilities, real-time order tracking, and one-tap approvals. Customer feedback turned from ridicule to praise. Mobile satisfaction scores jumped by 260%, and app adoption among clients rose from 18% to 87% within six months.

While tech improved internally, externally Nexora still lagged in how it communicated. Sales and marketing lacked social listening tools. CRM data was stale, and response times were measured in days. That changed with a Salesforce overhaul integrated with social platforms. AI sentiment analysis flagged client dissatisfaction in real time, triggering automated alerts and pre-drafted responses. One logistics director tweeted about poor onboarding—within two hours, Nexora had apologised, fixed the issue, and won public praise for responsiveness.

Challenges included adapting to differing social norms across regions and training teams to interpret AI outputs wisely ("No, the bot didn't mean we should DM that angry client a meme," became a mantra).

Even as transformation matured, Nexora kept one eye on the future. The sustainability team implemented e-waste audits, partnering with recyclers and moving 78% of workloads to carbon-neutral cloud providers. ISO 50001 certification soon followed.

Simultaneously, the company explored post-quantum cryptography to secure future data transmissions. Digital twins of warehouses were modelled using Unity and IoT data, optimising layout without moving a single box. Edge-cloud hybrid compute enabled faster decision loops. These initiatives weren't just PR. Clients with sustainability mandates began requesting Nexora's services over less green competitors, proving that future-proofing isn't just visionary—it's profitable.

Nexora's transformation under Project Helix was not a miracle—it was methodical, structured, and at times, messy. Guided by the ISLC at every turn, the company moved from firefighting to future-shaping. From outages to observability, from silos to self-service, from compliance chaos to competitive differentiation, Nexora emerged not just digitally stronger but culturally revitalised.

It now operates not on a rigid five-year IT plan, but on a living lifecycle, continuously evolving with each wave of change. The journey proved that transformation is not a project—it's a mindset. And perhaps most importantly, that enterprise IT doesn't have to be dull, robotic, or feared. With the right people, principles, and a bit of humour, even the most complex digital evolutions can be deeply human—and deeply successful.

	Sources: composed and written by EzraTech Ltd for the purpose of education in IT Systems Evaluation.
Essay questions (Choose one)	Each question requires you to analyse Nexora Group’s digital transformation and critically evaluate the organisation’s approach to information systems management. Your chosen essay should connect directly to the case study while being supported by wider research and theory. Q1. Generative AI & Agentic Intelligence: “Critically assess Nexora’s attempt to operationalise its generative AI system (RouteGen). Using the ISLC framework, evaluate how the organisation addressed infrastructure, governance, and cultural readiness challenges. What solutions would you propose to ensure long-term scalability of generative AI in Nexora’s logistics operations?” Q2. Data-Driven Operations: Democratization, Mesh, & Real-Time Analytics: “Analyse Nexora’s adoption of a data mesh approach to overcome siloed data. How effectively did the ISLC phases support the transition from fragmented data practices to real-time analytics? What additional system design or governance mechanisms would you recommend to balance data democratization with control?” Q3. Autonomy & Human–Machine Collaboration: “Evaluate Nexora’s introduction of cobots and multimodal interfaces in its warehouses. Using the ISLC, analyse how the company designed these systems to support human–machine collaboration rather than replacement. What risks and limitations should Nexora address to ensure long-term value from autonomous systems?” Q4. Governance, Sovereignty & Compliance: “Discuss how Nexora responded to global compliance pressures such as GDPR and DORA through sovereign cloud deployments and governance frameworks. Using the ISLC, critically assess whether its approach is sustainable across multiple regions. What improvements or alternative compliance strategies would you propose?” Q5. Enterprise Secure Access & Reducing Tech Debt: “Assess Nexora’s hybrid cloud migration as a strategy to reduce technical debt and improve secure access. Apply the ISLC to evaluate the success of its modular, microservices-based design. What additional measures could be introduced to further modernise Nexora’s infrastructure while reducing future debt accumulation?” Q6. AI in IT Ops: AIOps & Automated IT Support: “Investigate how Nexora could extend its AI adoption beyond logistics optimisation into IT operations (AIOps). Using the ISLC, evaluate the potential benefits and challenges of embedding AI-driven observability and automated IT support within its hybrid infrastructure. What design features should be prioritised?” Q7. Mitigating Data Sprawl & Strengthening Data Governance: “Critically evaluate Nexora’s shift towards automated data governance in response to AI-driven data growth. How well did the ISLC phases prepare the organisation to mitigate data sprawl and strengthen governance? Propose an enhanced system architecture that balances scalability with security.”

	Q8. Green Computing & Sustainable IT: “Analyse Nexora’s green IT initiatives, including cloud sustainability partnerships and e-waste audits. Using the ISLC, assess how sustainability principles were integrated into system design and operations. What further IS strategies could Nexora adopt to align with global sustainability standards while maintaining efficiency?” Q9. Emerging Technologies: Post-Quantum, Spatial, Hybrid Compute: “Examine Nexora’s exploration of emerging technologies such as post-quantum cryptography, spatial computing, and hybrid compute. Using the ISLC, evaluate how these innovations could future-proof Nexora’s IT systems. What risks, costs, and limitations should Nexora’s leadership anticipate in adopting such technologies?” <i>Students will achieve higher marks by demonstrating clear connection and application of relevant IS theories, and if they support their work with credible sources and provide appropriate citation and referencing.</i> <i>Sources: composed and written by EzraTech Ltd for the purpose of education in IT Systems Evaluation.</i>
Referencing:	Harvard Referencing should be used, see your Library Subject Guide for guides and tips on referencing: https://libguides.lsbu.ac.uk/ld.php?content_id=32487069
Regulations:	Make sure you understand the University Regulations on expected academic practice and academic misconduct. Note in particular: - Your work must be your own. Markers will be attentive to both the plausibility of the sources provided as well as the consistency and approach to writing of the work. Simply, if you do the research and reading, and then write it up on your own, giving the reference to sources, you will approach the work in the appropriate way and will not give markers reason to question the authenticity of the work. - All quotations must be credited and properly referenced. Paraphrasing is still regarded as plagiarism if you fail to acknowledge the source for the ideas being expressed. TURNITIN: TurnItIn will be enabled and will check for plagiarism. When you upload your work to the Moodle site it will be checked by anti-plagiarism software. - The usual assignment regulations, available on the Student VLE Portal, apply in respect of late submission penalties. - Copying someone else’s work, copying chunks of text directly out of a book or cutting and pasting from a web page is plagiarism and will be treated as an assessment offence by the university.

Learning Outcomes

This assessment (CW2 in the table below) will fully or partially assess the following learning outcomes for this module.

Learning outcome	Assessed by	
	CW1	CW2
A. LO1: Knowledge and Understanding		
• Understand the role of computer-based IS and how they support the business processes within an organisation	Fully	Partially
B. LO2: Intellectual Skills		
• Analyse organisational problems and engage in critical	Partially	Fully

evaluation of IS effectiveness		
C. LO3: Practical Skills		
• Develop a proposal for an IS that effectively supports organisational processes and decision making	Partially	Fully
D. LO4: Transferable Skills		
• Research skills, communication skills, writing skills, documentation skills, presentation skills, analytics skills, and being able to make recommendations	Partially	Fully

Assessment Criteria and Weighting

LSBU marking criteria have been developed to help tutors give you clear and helpful feedback on your work. They will be applied to your work to help you understand what you have accomplished, how any mark given was arrived at, and how you can improve your work in future.

How to get help

We will discuss this Coursework specification in class. However, if you have related questions, please ask a question on the general chat or if too specific contact your lecturer, Francis Babayemi (babayef3@lsbu.ac.uk) as soon as possible.

Quality assurance of coursework specifications

Coursework specifications within the CSI division go through internal (for new modules with 100% coursework also through external) moderation. This is to ensure high quality, consistency and appropriateness of the coursework as well as to share best practice within the CSI division.

Details of the moderators for this coursework specification are below:

Moderated (internal)	
Moderated (CSI lead)	
Signed off by (HoD/DHoD)	
-----For Internal use by CSI lead only-----	
Changes required to CW?	
Examples of good practice	

* if changes are required, moderator to complete the below:

List of changes required	
ML Response	
Moderator Response	

Module Title: Information Systems Management
Module Code: CSI_5_ISM
Assessment 2: Individual IS Evaluation Report Submission
Submission Date: Friday 19/12/25 - 5pm

Student ID Number / Participant ID

Assessment Criteria	Marks	Assessment Feedback
Contextualise the Challenge (10 marks): - Demonstrates understanding of Nexora's case study context. - Accurately identifies the relevant IS problem. - Explains why the challenge is significant to the organisation.	10.0	
Identify and Frame the Challenge (10 marks): - Defines the organisational/technological problem in ISM terms. - Clearly articulates the scope of the challenge. - Frames issue with reference to IS theory and case study.	10.0	
Conduct Evidence-Based Research (15 marks): - Uses recent, peer-reviewed academic sources (10–12 minimum). - Integrates industry reports and relevant case studies. - Demonstrates ability to synthesise and critique sources.	15.0	
Apply IS Frameworks (ISLC & others) (20 marks): - Sophisticated use of ISLC to analyse Nexora's challenge. - Correct mapping of ISLC stages to case study issues. - Application of additional frameworks (e.g., data governance models, AI ethics standards). - Demonstrates depth of theoretical integration.	20.0	
Critically Assess Solutions (with System Architecture Illustration) (20 marks): - Proposes realistic solutions linked to IS frameworks and case study. - Evaluates opportunities, risks, governance, infrastructure, and cultural readiness. - Includes at least one meaningful system architecture diagram/illustration. - Demonstrates strong connection between theory and practical solutions.	20.0	
Acknowledge Limitations (10 marks): - Recognises risks, constraints, and trade-offs. - Provides evidence of critical reflection (scalability, ethics, financial, cultural). - Discusses mitigation of limitations.	10.0	
Demonstrate Depth and Professionalism (15 marks): - Shows depth of analysis and integration of theory with practice. - Essay structure is coherent, logical, and professional. - Uses Harvard referencing correctly throughout. - Academic integrity upheld (no plagiarism, proper citations). - Writing is error-free in grammar and spelling. - Includes Cover Page with student credentials, module code, and submission date.	15.0	
Total	100.0	